Section 0: 8/25/25

Recall,

- $\mathbb{N} = \{1, 2, 3, \dots\}$.
- $\mathbb{Z} = \{\dots, -1, 0, 1, \dots\}$.
- $\mathbb{Q} = \{a/b \mid a, b \in \mathbb{Z}\}$.
- $\mathbb{R}$ = real numbers.

Section 1.5:

Def'n: Let $E \subseteq \mathbb{R}$. We call $M \in \mathbb{R}$ an <u>upper bound</u> for E (or say, E is bounded above by M) if for all $x \in E$, $x \leq M$.

Def'n: Let $E \subseteq \mathbb{R}$. We call $m \in \mathbb{R}$ a <u>lower bound</u> for E (or say, E is bounded below by m) if for all $x \in E$, $x \geq m$

If E is bounded above and below, we say that E is <u>bounded.</u>

Otherwise, E is <u>unbounded.</u>

Def'n: Let $E \subseteq \mathbb{R}$. If $M \in \mathbb{R}$ is an upperbound for E and $M \in E$, then M is called the maximum of E, ($\max E$.)

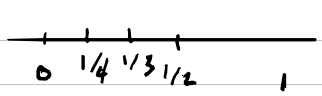
Ex:



$E = [0, 1]$. $\max E = 1$, $\min E = 0$.



$E = (0, 1)$.



$E = \{1/n \mid n \in \mathbb{N}\}$.
$\max E = 1$.

Section 1.6: 8/27/15

Def'n: Let $E \subseteq \mathbb{R}$, $E \neq \emptyset$. Assume E is bounded above. If $M \in \mathbb{R}$ satisfies:

1) M is an upper bound of E.

2) M is the least upper bound.
$\forall x \in \{\text{upper bounds of } E\}$, $M \leq x$.

Then M is the supremum of E. ($\sup E$)

Def'n: Let $E \subseteq \mathbb{R}$, $E \neq \emptyset$. Assume E is bounded below. If $m \in \mathbb{R}$ satisfies:

1) m is a lower bound of E.

2) m is the greatest lower bound.
$\forall x \in \{\text{lower bounds of } E\}$, $m \geq x$.

Then m is the infimum of E. ($\inf E$)

For convenience, we define:

- $\inf \emptyset = \infty$, $\sup \emptyset = -\infty$.
- If E is unbounded $\sup E = \infty$, $\inf E = -\infty$.

Def'n: Axiom of Completeness:

If $E \subseteq \mathbb{R}$, $E \neq \emptyset$ and E is bounded above. Then $\sup E$ exists and $\sup E \in \mathbb{R}$.

Section 1.7:

Theorem: (Archimedian Property)

The set $\mathbb{N}$ does not have an upper bound.

Proof: (by contradiction)
Assume φ is an upper bound for $\mathbb{N}$.
Since $\mathbb{N} \subseteq \mathbb{R}$ and $\mathbb{N} \neq \emptyset$, we can apply the axiom of completeness.

$\sup \mathbb{N} = x$.

Thus $\forall n \in \mathbb{N}$, $n \leq x$.

Because $x = \sup \mathbb{N}$, $x - 1$ is not an upper bound for $\mathbb{N}$.

Thus $\exists\, n \in \mathbb{N}$ s.t. $x - 1 < n$.

$\therefore x < n + 1 \Rightarrow x$ is not an upper bound.

Q.E.D.

Corollaries:

i) If $x \in \mathbb{R}$, then $\exists n \in \mathbb{N}$ s.t. $x < n$.

ii) If $x, y > 0$, then $\exists n \in \mathbb{N}$ s.t. $nx > y$.

iii) If $x > 0$, then $\exists n \in \mathbb{N}$ s.t. $\frac{1}{n} < x$.

Section 1.8:

Theorem: Well ordering property of $\mathbb{N}$.

Every nonempty subset of $\mathbb{N}$ has a smallest element.

Proof: Let $S \subseteq \mathbb{N}$ and $S \neq \emptyset$.

The axiom of completeness for infimums applies because S is bounded below by 1.

Let $\alpha = \inf S$. If $\alpha \in S$ then $\alpha = \min S$. So, assume $\alpha \notin S$.

Then there must exist $x \in S$ s.t. $x < \alpha + 1$.

But $\alpha = \inf S$ and $\alpha \neq x$, so $\alpha < x$.

$\therefore \alpha < x < \alpha + 1$.

x is not a lower bound for S, so there must exist $y \in S$ s.t. $y < x$.

$\therefore \alpha < y < x < \alpha + 1$.

$\therefore |x - y| < 1$.

but this is impossible, because $x - y \in \mathbb{N}$, and no natural # can be less than 1.

Contradiction!

$\Rightarrow$ Every nonempty subset of $\mathbb{N}$ has a least element.

Q.E.D.

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Section 1.9: 9/03/25

Def'n: A set $E \subseteq \mathbb{R}$ is said to be dense in $\mathbb{R}$ if for every nonempty interval (x,y), there exists $c \in E$ s.t. $c \in (x,y)$.

$(E \cap (x,y) \neq \varnothing)$.

Theorem: The set $\mathbb{Q}$ is dense in $\mathbb{R}$.

Proof: Let $(x,y) \subseteq \mathbb{R}$. $(x < y)$

$\exists n \in \mathbb{N}$ s.t. $\frac{1}{n} < y - x$. (Hw 1 2c)

Now consider $E = \{k \in \mathbb{Z} : \frac{k}{n} > x\}$.
(It can be shown by the well ordering principle, that E is bounded below).

Let $m = \min(E)$.

So, $\frac{m-1}{n} \leq x < \frac{m}{n}$.

Because $\frac{m-1}{n} \leq x$, then $\frac{m}{n} - \frac{1}{n} \leq x$ so.

$\frac{m}{n} \leq x + \frac{1}{n} < x + (y - x)$

$\Rightarrow x < \frac{m}{n} \leq x + \frac{1}{n} < y$

$\Rightarrow x < \frac{m}{n} < y$.

Q.E.D.

Section 1.10:

Def'n: For $x \in \mathbb{R}$, $|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$.

Theorem: Let $x, y \in \mathbb{R}$. Then

i) $-|x| \leq x \leq |x|$.

ii) $|xy| = |x| \cdot |y|$.

iii) $|x+y| \leq |x| + |y|$.

iv) $||x| - |y|| \leq |x - y|$.

Proof (The triangle inequality):

C_1: Let $x, y \geq 0$.

$|x+y| = x + y = |x| + |y|$.

C_2: $x, y < 0$.

$|x+y| = -(x+y) = -x - y = |x| + |y|$.

C_3: $x \geq 0$, $y < 0$ without loss of generality.

SC_{3a}: $x + y \geq 0$

$|x+y| = x + y \leq |x| + |y|$. $(= x - y)$.

SC_{3b}: $x + y \leq 0$.

$|x+y| = -(x+y) = -x - y = -x + |y|$

and $-x + |y| \leq |x| + |y|$.

$\therefore |x+y| \leq |x| + |y|$.

Q.E.D.

Def'n: We define the Euclidean distance function on $\mathbb{R}$ by $d(x,y) = |x - y|$, $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$.

Any distance function satisfies:

1) $d(x,y) \geq 0$.

2) $d(x,y) = 0 \Leftrightarrow x = y$.

3) $d(x,y) = d(y,x)$.

4) $d(x,y) \leq d(x,z) + d(z,y)$.

Euclidean distance satisfies because:

1) $|x - y| \geq 0$.

2) $|x - y| = 0 \Rightarrow x - y = 0 \Rightarrow x = y$.

3) $|x - y| = |-(x - y)| = |y - x|$.

4) $|x - y| = |x - z + z - y| = |(x - z) + (y - z)|$

$\leq |x - z| + |z - y|$.

Section 2.2: 9/08/25

Def'n: A sequence of real numbers is a function:

$$f: \mathbb{N} \to \mathbb{R}.$$

Def'n: An arithmetic sequence has the form:

$$c, c+d, c+2d, c+3d, \dots$$

explicit: $x_n = c+(n-1)d$, $n \geq 1$.

implicit: $x_1 = c$, $x_n = x_{n-1} + d$.

Def'n: A geometric sequence has the form:

$$c, cr, cr^2, cr^3, \dots$$

explicit $x_n = cr^{n-1}$, $n \in \mathbb{N}$.

Def'n: If $f: A \to \mathbb{R}$ is any function we can create an iterated sequence by picking $c \in A$ and setting

Exp: $x_1 = c$, $x_2 = f(c)$, $x_3 = f \circ f(c)$, ...

imp: $x_1 = c$, $x_n = f(x_{n-1})$.

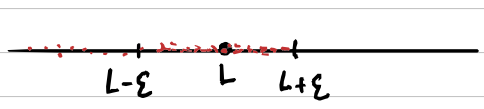
Section 2.4:

Def'n: Let $\{S_n\}$ be a sequence in $\mathbb{R}$. We say that $\{S_n\}$ converges to L and write $\lim S_n = L$ if for every $\varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $|S_n - L| < \varepsilon$ $\forall n \geq N$.

Alternative Notation:

$S_n \to L$ as $n \to \infty$ instead of $\lim_{n\to\infty} S_n = L$.

Idea:



All S_n with $n \geq N$ are inside $(L-\varepsilon, \varepsilon+L)$.

Ex: Prove that $\lim_{n\to\infty} n^2/(2n^2+1) = \frac{1}{2}$.

Proof: Let $\varepsilon > 0$. We must find $N \in \mathbb{N}$ s.t. $n > N \Rightarrow \left|\frac{n^2}{2n^2+1} - \frac{1}{2}\right| < \varepsilon$.

Observe that $\frac{n^2}{2n^2+1} - \frac{1}{2} = \frac{2n^2-(2n^2+1)}{4n^2+2} = \frac{-1}{4n^2+2}$.

$\Rightarrow \left|\frac{-1}{4n^2+2}\right| = \frac{1}{4n^2+2}$.

So, $\frac{1}{4n^2+2} < \varepsilon \Rightarrow 1 < 4n^2\varepsilon + 2\varepsilon \Rightarrow \frac{1}{\varepsilon} < 4n^2 + 2$.

$\therefore 4n^2 > \frac{1}{\varepsilon} - 2 \Rightarrow n^2 > \frac{1}{4}\left(\frac{1}{\varepsilon} - 2\right) \Rightarrow n > \frac{1}{2}\sqrt{\frac{1}{\varepsilon} - 2}$.

So, let $N = \frac{1}{2}\sqrt{\frac{1}{\varepsilon} - 2}$.

(By the Archimedean Principle, there exists $N \in \mathbb{N}$ s.t. $N > \frac{1}{2\sqrt{\varepsilon}}$. So if $n > N$, then $\left|\frac{n^2}{2n^2+1} - \frac{1}{2}\right| < \varepsilon$ is satisfied.)

Hence, $\lim_{n\to\infty} \frac{n^2}{2n^2+1} \to \frac{1}{2}$.

Q.E.D.

Theorem: (uniqueness of limits) Suppose that $\lim_{n\to\infty} S_n = L_1$ and $\lim_{n\to\infty} S_n = L_2$. Then $L_1 = L_2$.

Proof: Let $\varepsilon > 0$. Then because $\lim S_n = L_1$, then $\exists N_1 \in \mathbb{N}$ s.t. $|S_n - L_1| < \varepsilon$ $\forall n > N_1$, and $\exists N_2 \in \mathbb{N}$ s.t. $|S_n - L_2| < \varepsilon$ $\forall n > N_2$.

Let $m = \max\{N_1, N_2\}$. Then both $|S_m - L_1| < \varepsilon$ and $|S_m - L_2| < \varepsilon$.

Then $|L_1 - L_2| = |L_1 - S_m + S_m - L_2| \leq |L_1 - S_m| + |L_2 - S_m| < 2\varepsilon$.

Thus $|L_1 - L_2| < 2\varepsilon$ $\forall \varepsilon > 0$.

Thus $L_1 - L_2 = 0$.

Q.E.D.

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Section 2.5: 9/15/25

Def'n: Let $\{S_n\}$ be a sequence of real numbers.

$\{S_n\}$ diverges to ∞ and write $\lim_{n\to\infty} S_n = \infty$

if $\forall M > 0, \exists N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow S_n > M$.

Ex: Show that $\lim_{n\to\infty} \frac{n^2+1}{n+1} = \infty$.

Proof: Let $M > 0$. Solve $\frac{n^2+1}{n+1} \geq M$.

observe: $\frac{n^2+1}{n+1} = \frac{(n^2+2n+1)-2n}{n+1} = \frac{(n+1)^2-2n}{n+1}$

$= n+1 - \frac{2n}{n+1}$. (completing the square)

So, $n+1 - 2\left(\frac{n}{n+1}\right) \geq M \Rightarrow n \geq M + \frac{2n}{n+1} - 1$

hmm...

$M + \frac{2n}{n+1} - 1 \geq M + 2 - 1 = M+1$.

So, $n \geq M+1$.

Let $N = M+1$. Then, if $n \geq N$, $\frac{n^2+1}{n+1} \geq M$.

Section 2.6:

Def'n: A sequence $\{S_n\}$ is bounded if there exists $M \in \mathbb{R}$ s.t. $|S_n| < M$ $\forall n$.

Theorem: Every convergent sequence is bounded.

Proof: Assume $S_n \to L$ as $n \to \infty$. $(L \in \mathbb{R})$.

By definition of convergence,

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$ s.t. $|S_n - L| < \varepsilon$ $\forall n > N$.

Let $\varepsilon = 1$. So, $\exists N_1$ s.t. $\forall n > N_1$, $|S_n - L| < 1$.

Observe that $|S_n| = |S_n - L + L| \leq |S_n - L| + |L|$.

$\Rightarrow |S_n - L| + |L| < 1 + |L|$ if $n > N_1$.

Let $M = \max\{|L|+1, |S_1|, |S_2|, \ldots, |S_{N_1}|, 1\}$.

Then, for any $n \in \mathbb{N}$, $|S_n| \leq M$.

Q.E.D.

Section 2.7:

Theorem: Assume $\{S_n\}$ is convergent and $c \in \mathbb{R}$.

Then $\lim_{n\to\infty} c \cdot S_n = c \cdot \lim_{n\to\infty} S_n$.

Proof: Let $L = \lim_{n\to\infty} S_n$. We want to prove that $\lim_{n\to\infty} c \cdot S_n = c \cdot \lim_{n\to\infty} S_n = cL$.

So, we want $|c \cdot S_n - cL| < \varepsilon$.

$\Rightarrow |c| \cdot |S_n - L| < \varepsilon \Rightarrow |S_n - L| < \frac{\varepsilon}{|c|}$. If $c \neq 0$.

Let $\varepsilon > 0$ and let $c \neq 0$. Pick $N \in \mathbb{N}$ s.t. $|S_n - L| < \frac{\varepsilon}{|c|}$ $\forall n > N$.

Then, $|c||S_n - L| < \varepsilon \Rightarrow |cS_n - cL| < \varepsilon$.

Therefore, $\lim_{n\to\infty} c \cdot S_n = cL$.

Q.E.D.

Theorem: If $\{S_n\}$ and $\{t_n\}$ are convergent, then

$\lim_{n\to\infty} (S_n + t_n) = \lim_{n\to\infty} S_n + \lim_{n\to\infty} t_n$.

Theorem: If $\{S_n\}$ and $\{t_n\}$ are convergent, then

$\lim_{n\to\infty} (S_n \cdot t_n) = \lim_{n\to\infty} S_n \cdot \lim_{n\to\infty} t_n$.

Proof: Let $\lim_{n\to\infty} S_n = S$ and $\lim_{n\to\infty} t_n = T$.

We need N s.t. $|S_n t_n - ST| < \varepsilon$ if $n > N$.

Observe $|S_n t_n - ST| = |S_n t_n - S_n T + S_n T - ST|$.

$\leq |S_n t_n - S_n T| + |S_n T - ST|$.

$= |S_n|(|t_n - T| + |T||S_n - S|$.

($|S_n|$: bounded by convergence; $|t_n - T|$: small by convergence; $|T|$: constant; $|S_n - S|$: small by convergence.)

$< \varepsilon/2$ $\qquad$ $< \varepsilon/2$

Now, we start the formal proof.

Since $\{S_n\}$ is convergent, it is bounded, so, $\exists M \in \mathbb{R}$ s.t. $|S_n| \leq M$ $\forall n$.

- By the convergence of $\{t_n\}$, there exists $N_1 \in \mathbb{N}$ s.t. if $n \geq N_1$, then

$|t_n - T| < \frac{\varepsilon}{2M}$.

- By convergence of $\{S_n\}$, there exists $N_2 \in \mathbb{N}$ s.t. $n > N_2$, then

$|S_n - S| < \frac{\varepsilon}{2|T|}$. To account for $|T| = 0$, we use:

$|S_n - S| < \frac{\varepsilon}{2 \cdot |T| + 2}$.

Let $N = \max\{N_1, N_2\}$. Then, if $n > N$,

$|S_n - S| < \frac{\varepsilon}{2|T|+2}$ and $|t_n - T| < \frac{\varepsilon}{2M}$.

$|S_n t_n - ST| \leq |S_n||t_n - T| + |T||S_n - S|$

$\leq M\left(\frac{\varepsilon}{2} \cdot \frac{1}{M}\right) + |T|\left(\frac{\varepsilon}{2} \cdot \frac{1}{|T|+1}\right)$

$= \frac{\varepsilon}{2} + \frac{\varepsilon}{2}\left(\frac{|T|}{|T|+1}\right)$

$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$.

Q.E.D.

Theorem: If $\{S_n\}$ and $\{t_n\}$ are convergent, $t_n \neq 0$ and $\lim_{n\to\infty} t_n \neq 0$, then:

$\lim_{n\to\infty} \frac{S_n}{t_n} = \frac{\lim_{n\to\infty} S_n}{\lim_{n\to\infty} t_n}$.

Section 2.8: 9/17/25

lemma: Suppose $\{s_n\}$ is convergent. If $s_n \geq 0$ for all $n \in \mathbb{N}$, then $\lim_{n\to\infty} s_n \geq 0$.

Proof: Let $S = \lim_{n\to\infty} s_n$. We want to prove that $S \geq 0$.

Suppose for contradiction that $S < 0$.

Let $\varepsilon = |S|$. Then by convergence,

$\exists N \in \mathbb{N}$ s.t. $\forall n \geq N$, $|s_n - S| < |S|$.

In particular, $|s_N - S| < |S|$.

Now, $S < 0$ so $|S| = -S$, and $s_N \geq 0$

So, $s_N - S > 0$, hence $|s_N - S| = s_N - S$.

$s_N - S < -S \Rightarrow s_N < 0$. Contradiction!

This contradicts $s_N \geq 0$.

Q.E.D.

Theorem: (order limit theorem) Suppose $\{s_n\}$ and $\{t_n\}$ are convergent with $\{s_n\} \leq \{t_n\}$.

Then $\lim_{n\to\infty} s_n \leq \lim_{n\to\infty} t_n$.

Proof: Because $s_n \leq t_n$, then $0 \leq t_n - s_n$. apply the above lemma to $\{t_n - s_n\}$ gives

$$0 \leq \lim_{n\to\infty}(t_n - s_n)$$

$$\Rightarrow 0 \leq \lim_{n\to\infty} t_n - \lim_{n\to\infty} s_n$$

$$\Rightarrow \lim_{n\to\infty} s_n \leq \lim_{n\to\infty} t_n.$$

Q.E.D.

Note: If $s_n < t_n$ $\forall n$, it is not necessarily true.

Theorem: (The squeeze theorem) Let $\{s_n\}, \{x_n\}, \{t_n\}$ be sequences with $L = \lim_{n\to\infty} s_n = \lim_{n\to\infty} t_n$. and $s_n \leq x_n \leq t_n$ $\forall n$. Then $\{x_n\}$ converges to L.

Proof: Let $\varepsilon > 0$. By assumption, there exists $N_1 \in \mathbb{N}$ s.t. if $n \geq N_1$, then $|s_n - L| < \varepsilon$, hence $-\varepsilon < s_n - L < \varepsilon$.

there exists $N_2 \in \mathbb{N}$ s.t. if $n \geq N_2$, then $|t_n - L| < \varepsilon$, and thus $-\varepsilon < t_n - L < \varepsilon$.

Let $N = \max\{N_1, N_2\}$. So, for $n \geq N$,

$$s_n \leq x_n \leq t_n$$

$$\Rightarrow s_n - L \leq x_n - L \leq t_n - L.$$

$$\Rightarrow -\varepsilon < s_n - L \leq x_n - L \leq t_n - L < \varepsilon.$$

$$\Rightarrow -\varepsilon < x_n - L < \varepsilon \Rightarrow |x_n - L| < \varepsilon.$$

Q.E.D.

Section 2.9:

Def'n: We say a sequence $\{s_n\}$ is

- Increasing if $s_n < s_{n+1}$ $\forall n \in \mathbb{N}$.
- Decreasing if $s_n > s_{n+1}$ $\forall n \in \mathbb{N}$.
- nondecreasing if $s_n \leq s_{n+1}$ $\forall n \in \mathbb{N}$.
- nonincreasing if $s_n \geq s_{n+1}$ $\forall n \in \mathbb{N}$.

If any of the 4 above are true, then the sequence is called monotone.

Theorem: (Monotone convergence theorem). If $\{s_n\}$ is bounded and monotone, then it converges.

Proof: Without loss of generality assume that s_n is nondecreasing. Consider the set $E = \{s_n : n \in \mathbb{N}\}$ by assumption, E is bounded above.

By the axiom of completeness, $S = \sup E$. We claim $\lim_{n\to\infty} s_n = S$.

Let $\varepsilon > 0$. Because $S = \sup E$, then $S - \varepsilon$ is not an upper bound for E.

$\Rightarrow \exists s_N \in E$ s.t. $S - \varepsilon < s_N$.

Because s_n is nondecreasing, for all $n \geq N$, $s_n \geq s_N$.

So, $S - \varepsilon < s_N \leq s_n \leq S < S + \varepsilon$.

$\Rightarrow \forall n > N$, $|s_n - S| < \varepsilon$.

Q.E.D.

Section 2.9: 9/22/25

Ex: Consider $\{a_n\}$, defined by $\sqrt{2}, \sqrt{2+\sqrt{2}}, \ldots$

$a_1 = \sqrt{2}$, $a_n = \sqrt{2+a_{n-1}}$.

What is $\lim_{n\to\infty} a_n$? We will use MCT.

Claim: $a_{n-1} < a_n$.

Proof: $a_1 = \sqrt{2}$, $a_2 = \sqrt{2+\sqrt{2}}$, $\Rightarrow a_1 < a_2$.

Suppose $\exists m \in \mathbb{N}$, s.t. $a_m < a_{m+1}$.

$\Rightarrow a_m + 2 < a_{m+1} + 2$.

$\Rightarrow \sqrt{a_m+2} < \sqrt{a_{m+1}+2}$.

$\Rightarrow a_{m+1} < a_{m+2}$.

Q.E.D.

claim: $\{a_n\}$ is bounded

Proof: $a_1 = \sqrt{2} < 4$.

Suppose $\exists m \in \mathbb{N}$ s.t. $a_m < 4$.

$\Rightarrow a_m + 2 < 6 \Rightarrow \sqrt{2+a_m} < \sqrt{6} < 4$.

$\Rightarrow a_{m+1} < 4$.

$\therefore \{a_n\}$ is bounded above by 4.

Q.E.D.

We invoke the monotone convergence theorem to state that $\{a_n\}$ is convergent.

Let $L = \lim_{n\to\infty} a_n$.

$x_n = \sqrt{2+x_{n-1}}$. $x_n^2 = 2 + x_{n-1}$.

$\lim_{n\to\infty} a_n^2 = \lim_{n\to\infty}(2 + a_{n-1})$

$\Rightarrow L^2 = 2 + L. \Rightarrow L^2 - L - 2 = 0.$

$\Rightarrow (L-2)(L+1)$.

$L = 2$ or $L = -1$.

$L = 2$ by es obvio que.

(or because $a_n > 0$ $\forall n$ and $L = 0$).

Section 2.11:

Def'n: Let $\{s_n\}$. A subsequence of $\{s_n\}$ is any sequence $\{s_{n_k}\} = \{s_{n_1}, s_{n_2}, \ldots\}$ where $n_1 < n_2 < \ldots$

Ex: $\{s_3, s_5, s_7, s_{101}, \ldots\}$.

Another Ex: Let $\{s_n\} = \{1, 2, 3, \ldots\}$.
Then $\{2, 4, 8, \ldots\}$ is a subsequence and $\{1, 3, 5, \ldots\}$ is a subsequence.

Theorem: Every sequence has a monotonic subsequence.

Proof: Let $\{x_n\}$ be a sequence.

Call an element x_m a turnback point if $x_m \geq x_n$ $\forall n > m$.

1. If there are infinitely many turnback points $x_{m_1}, x_{m_2}, \ldots$ then $\{x_{m_1}, x_{m_2}, x_{m_3}, \ldots\}$ is non-increasing because $x_{m_1} \geq x_{m_2} \geq x_{m_3} \geq \ldots$

2. If there are finitely many turnback points then let x_M be the last one.
(If there are none, use x_1).
Then x_{M+1} is not a turnback point. Hence, there exists x_{m_1}, $m_1 > M+1$ s.t. $x_{m_1} > x_{M+1}$.
Similarly, x_{m_1} is not a turnback point. So there exists x_{m_2} with $m_1 < m_2$ and $x_{m_2} > x_{m_1}$.

Then $\{x_{M+1}, x_{m_1}, x_{m_2}, x_{m_3}, \ldots\}$ is an increasing sequence.

Q.E.D.

Theorem: Bolzano-Weierstrass theorem.
Every bounded sequence has a convergent subsequence.

Proof: Assume $\{x_n\}$ is bounded.
By last one, it has a monotone subsequence.
(which is also bounded)
By MCT, we're done.

Q.E.D.

Ex: $\{-1, 1, -1, 1, -1, 1, \ldots\} = \{-1^n\}$.

Section 2.12:

Def'n: A sequence $\{s_n\}$ is called cauchy if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. if $m, n \geq N$, then $|s_n - s_m| < \varepsilon$.

Section 2.12: 9/24/25

Def'n: A sequence $\{s_n\}$ is called cauchy if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. if $m, n \geq N$, then $|s_n - s_m| < \varepsilon$.

Lemma: If a sequence is Cauchy, then it is bounded.

Proof: Left as an exercise to the reader. Shucks.

Theorem: A sequence $\{s_n\}$ is convergent iff it is cauchy.

Proof: ($\Rightarrow$) Suppose $\{s_n\}$ is convergent. Let $\varepsilon > 0$. Then, by convergence, there exists $N \in \mathbb{N}$ s.t. if $k \geq N$, then $|s_k - L| < \varepsilon/2$.

Let $m, n \geq N$. Then
$$|s_n - s_m| = |s_n - L - (s_m - L)|$$
$$\leq |s_n - L| + |s_m - L|$$
$$< \varepsilon/2 + \varepsilon/2 = \varepsilon.$$
$$\Rightarrow |s_n - s_m| < \varepsilon. \square$$

($\Leftarrow$) Suppose $\{s_n\}$ is Cauchy. By the above lemma, $\{s_n\}$ is bounded. By the Bolzano-Weierstrass theorem, there exists $\{s_{n_k}\}$ which is convergent to L. We claim that $\{s_n\}$ converges to L as well.

Let $\varepsilon > 0$, then $\exists N \in \mathbb{N}$ s.t. $|s_n - s_m| < \varepsilon/2$ for $n, m \geq N$.

Because $\{s_{n_k}\}$ converges to L, $\exists K \in \mathbb{N}$ s.t. if $k \geq N$, then $|s_{n_k} - L| < \varepsilon/2$.

Let $k \in \mathbb{N}$ s.t. $n_k \geq N$, then

$$|s_n - L| = |s_n - s_{n_k} + s_{n_k} - L|$$
$$\leq |s_n - s_{n_k}| + |s_{n_k} - L|$$
$$< \varepsilon/2 + \varepsilon/2 = \varepsilon. \square$$

$\therefore$ Cauchy $\Leftrightarrow$ Convergence.

Q.E.D.

Ex: Let $x_1 = 1$, $x_2 = 2$, $x_n = \frac{x_{n-1} + x_{n-2}}{2}$.

Note that if $n, m \geq N$, then $|x_n - x_m| \leq \frac{1}{2^{N-1}}$.

Clearly, the sequence is cauchy, hence, it converges.

Chapter 3 (Briefly):

Def'n: Let $\{a_k\}$ be a sequence. The nth partial sum is
$$s_n = \sum_{k=1}^{n} a_k.$$

If s_n is convergent, then $\lim_{n\to\infty} s_n = \sum_{k=1}^{\infty} a_k$.

Comparison Test: Suppose $\{a_k\}$ and $\{b_k\}$ are sequences with $0 \leq a_k \leq b_k$. Then:

i) If $\sum_{k=1}^{\infty} b_k$ converges, then $\sum_{k=1}^{\infty} a_n$ converges.

ii) If $\sum_{k=1}^{\infty} a_k$ diverges, then $\sum_{k=1}^{\infty} b_k$ diverges.

Proof: Let $s_n = \sum_{k=1}^{n} a_k$ and $t_n = \sum_{k=1}^{n} b_k$

i) Suppose that $\{t_n\}$ converges. Thus, $\{t_n\}$ is bounded.

Because $s_n \leq t_n$, then s_n is bounded. By def'n, s_n is nondecreasing.

By MCT, $\{s_n\}$ converges. $\square$

ii) Suppose that $\{s_n\}$ diverges.

So for any $M \in \mathbb{R}$, $\exists N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow s_n \geq M$. Because $t_n \geq s_n$, $M \leq s_n \leq t_n \Rightarrow t_n \geq M$, so $\{t_n\}$ diverges to ∞. $\square$

Q.E.D.

Ex: Geometric Series.

Let $r, a \in \mathbb{R}$. $(r \neq 1)$. A geometric series is of the form:
$$\sum_{k=0}^{\infty} ar^k.$$

We can observe that $(1-r)(1 + r + r^2 + \cdots + r^{m-1}) = 1 - r^m$.

$$s_m = 1 + r + r^2 + \cdots + r^{m-1} = \frac{1 - r^m}{1 - r}.$$

So, by lim properties, $\lim_{m\to\infty} s_m = \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$, $|r| < 1$.

Section 4.2: 9/29/25

Def'n: Let $E \subseteq \mathbb{R}$. $x \in E$ is an <u>interior point of E</u> if there is $c \in \mathbb{R}$ s.t. $(x-c, x+c) \subseteq E$.

Ex:

i) Every point in (a,b) is an interior point.

ii) The endpoints of a closed interval $[a,b]$ are <u>not</u> interior points.

iii) $\mathbb{N}$ and $\mathbb{Q}$ have no interior points.

iv) Every point of $\mathbb{R}$ is an interior point.

Def'n: Let $E \subseteq \mathbb{R}$. We call $x \in E$ an <u>isolated point in E</u> if there is some $c \in \mathbb{R}$ s.t. $(x-c, x+c) \cap E = \{x\}$.

Ex:

i) Every point in $\mathbb{N}$ and $\mathbb{Z}$ is an isolated point.

ii) (a,b), $[a,b]$, $\mathbb{Q}$, $\mathbb{R}$ have no isolated points.

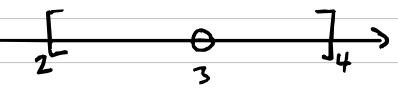
Def'n: Let $E \subseteq \mathbb{R}$. We call $x \in \mathbb{R}$ an <u>accumulation point</u> if $\forall c > 0$, $(x-c, x+c) \cap E \supsetneq \{x\}$.

Ex:

i) Every point of (a,b) is an accumulation point, along with a, and b.

ii) No point of $\mathbb{N}$ is an accumulation point.

iii) $[2,4] - \{3\}$.



Every point of $[2,4]$ is an accumulation point for E.

Proposition: Let $E \subseteq \mathbb{R}$. Then $x \in \mathbb{R}$ is an accumulation point of E iff there is a sequence $\{x_n\}$ with $x_n \in E$, and $x_n \neq x$ s.t. $\lim_{n\to\infty} x_n = x$.

Proof: ($\Rightarrow$) Assume $x \in \mathbb{R}$ is an accumulation point of E.

We construct a sequence as follows:

For $n=1$, the s.t. $(x-1, x+1)$ contains at least one point besides x. Let $x_1 \in (x-1, x+1) - \{x\}$, $(x_1 \in E)$.

Similarly, pick $x_n \in (x - \frac{1}{n}, x + \frac{1}{n})$ in the same way.

$\therefore x_n \in E$ and $|x - x_n| < \frac{1}{n}$.

So, if $\varepsilon > 0$, by the archimedean principle, $\exists n \in \mathbb{N}$ s.t. $\frac{1}{n} < \varepsilon$ and thus $|x - x_n| < \frac{1}{n} < \varepsilon \Rightarrow |x - x_n| < \varepsilon$. $\square$

($\Leftarrow$) Assume that $\lim_{n\to\infty} x_n = x$, where $x_n \in E$ and $x \neq x_n$. Let $c \in \mathbb{R}$. By convergence, there is $N \in \mathbb{N}$ s.t. $|x - x_N| < c$. So, $x_N \in (x-c, x+c) \cap E$ and $x_N \neq x$. Thus x is an accumulation point. $\square$

Q.E.D.

Def'n: Let $E \subseteq \mathbb{R}$. We call $x \in \mathbb{R}$ a <u>boundary point of E</u> if $\forall c > 0$, $(x-c, x+c)$ contains at least one point from E and at least one point that is not in E.

Ex:

i) a and b are boundary points in (a,b) and $[a,b]$.

ii) Every point of $\mathbb{N}$ is a boundary point for $\mathbb{N}$.

iii) $\mathbb{R}$ has no boundary points.

iv) Every point of $\mathbb{R}$ is a boundary point for $\mathbb{Q}$.

Section 4.3: 10/06/25

Def'n: Let $E \subseteq \mathbb{R}$. We call E closed if every accumulation point of E is in E.

Examples:
i) $\emptyset$
ii) (a,b)
iii) $[a,b]$
iv) $\mathbb{N}$
v) $\mathbb{R}$.

Def'n: Let $E \subseteq \mathbb{R}$. E is open if every point in E is also an interior point.

Def'n: Let $E \subseteq \mathbb{R}$. Let E' be the set of accumulation points of E. We call $\bar{E} = E \cup E'$ the closure of E.

Def'n: Let $E \subseteq \mathbb{R}$. The set of interior points, $\text{int}\, E$.
$\text{int}\, E = \{x \in E : x \text{ is an interior point}\}$.

$\text{int}\, E$ is called the interior of E.

Section 4.5:

Theorem: Let $A \subseteq \mathbb{R}$ and $B = \mathbb{R} - A$. Then A is open iff B is closed.

Proof: ($\Rightarrow$) By contradiction, suppose that A is open and B is not closed. So there exists an accumulation point z that is not in B, hence it is in A. Then z is an interior point for A. So there exists $(z-\delta, z+\delta)$ inside of A. $(z-\delta, z+\delta)$ contains no points from B, which contradicts it being an accumulation point of B. ■

($\Leftarrow$) Now assume B is closed and A is not open. Let z be a point of A that is not interior. So for any $\varepsilon > 0$, the interval $(z-\varepsilon, z+\varepsilon)$ contains some points of B. Let $\varepsilon = 1/n$, let $b_n \in (z - 1/n, z + 1/n) \cap B$, hence $\lim_{n\to\infty} b_n = z$. So z is an accumulation point of B. Because B is closed, $z \in B$. But it was assumed that $z \in A$. ■

Q.E.D.

Section 4.5: 10/08/25

Theorem: Properties of open sets.

i) The sets $\emptyset$ and $\mathbb{R}$ are open

ii) Any finite intersection of open sets is open.

iii) Any infinite union of open sets is open.

iv) The complement of an open set is closed.

Proof:

ii) Let's start with the case of two sets. Suppose E_1 and E_2 are open. We want to show that $E_1 \cap E_2$ is open. Let $z \in E_1 \cap E_2$. We wish to show that z is an interior point of $E_1 \cap E_2$. Because E_1 and E_2 are open, z is an interior point of them. Therefore $\exists (z-\delta_1, z+\delta_1) \subset E_1$ and $\exists (z-\delta_2, z+\delta_2) \subset E_2$. take $\delta = \min\{\delta_1, \delta_2\}$.
$\Rightarrow (z-\delta, z+\delta) \subset E_1$ and $(z-\delta, z+\delta) \subset E_2$
$\Rightarrow (z-\delta, z+\delta) \subset E_1 \cap E_2 \Rightarrow z$ is an interior point of $E_1 \cap E_2$.
$\therefore E_1 \cap E_2$ is open.
It follows by induction that $E_1 \cap E_2 \cap \dots \cap E_n$ is open. ■

iii) Consider $E = \bigcup_{n=1}^{\infty} E_n$, where each E_n is open. If $z \in E$, then $z \in E_j$, $j \in \mathbb{N}$. E_j is open so z is an interior point of E_j; hence, there is an interval $(z-\varepsilon, z+\varepsilon) \subseteq E_j \subseteq E$. So we are done. ■

Q.E.D.

Theorem: Properties of closed Sets.

i) $\emptyset$ and $\mathbb{R}$ are closed.

ii) Any finite union of closed sets is closed

iii) Any intersection of closed sets is closed.

iv) The complement of a closed set is open.

Proof:

ii) Let F_1 and F_2 be closed

$\overline{F_1 \cup F_2} = \bar{F}_1 \cap \bar{F}_2$ is open.
$\Rightarrow F_1 \cup F_2$ is closed. ■

iii) Is similar. □

Q.E.D.

Ex: $\bigcup_{n=2}^{\infty} [\frac{1}{n}, 1-\frac{1}{n}] = (0,1)$. ☹

$[0,1] \cup (0,2) = [0,2)$. ☹

Section 4.5:

Def'n: A set $k \subseteq \mathbb{R}$ is compact if every sequence in k has a subsequence that converges to a limit also in k.

Theorem: Heine-Borel.
A set is compact iff it is closed and bounded.

Proof: Suppose k is compact. Suppose for sake of contradiction that k is not bounded. So, there exists $x_1 \in k$ s.t. $|x_1| > 1$, $\exists x_2$ s.t. $|x_2| > 2, \dots$ $\exists x_n$ s.t. $|x_n| > n$. By compactness, there is a convergent subsequence $\{x_{n_k}\}$ of $\{x_n\}$. Because $\{x_n\}$ is bounded and monotone, $\{x_{n_k}\}$ is unbounded and monotone and thus divergent. ■

To prove that k is closed, let x be an accumulation point of k. So, there exists $\{x_n\}$ with $x_n \in k$ and $\lim_{n\to\infty} x_n = x$. By Compactness, there is a subsequence $\{x_{n_k}\}$ that is converging to a point in k. $\lim_{k\to\infty} x_{n_k} = x$, so $x \in k$. Thus k is closed. ■

($\Leftarrow$) Assume k is closed and bounded. Let $\{x_n\}$ be a sequence in k. Then $\{x_n\}$ is bounded, so it has a convergent subsequence (by Bolzano-Weierstrass theorem). So, $\lim_{k\to\infty} x_{n_k} = x$. Thus x is an accumulation point for k. by def'n of closure, $x \in k$. Thus, k is closed and bounded. ■

Q.E.D.

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Section Something

Section 5.1: 10/20

Def'n: (δ-ε) Let $f: E \to \mathbb{R}$ and let x_0 be an accumulation point of E and $L \in \mathbb{R}$. We write $\lim_{x \to x_0} f(x) = L$ if

$\forall \varepsilon > 0, \exists \delta > 0$ s.t. $|x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon$.

Ex: Prove that $\lim_{x \to 5} (10x - 11) = 39$.

Proof: Let $\varepsilon > 0$.

$$|x - 5| < \delta \Rightarrow |39 - 10x + 11| < \varepsilon.$$
$$|50 - 10x| < \varepsilon.$$
$$10|5 - x| < \varepsilon.$$

$$\therefore \delta < \varepsilon/10 \Rightarrow |x - 5| < \frac{\varepsilon}{10}$$

$$\Rightarrow 10|5 - x| < 10\frac{\varepsilon}{10} = \varepsilon.$$

Q.E.D.

Def'n: (Sequential) Let $f: E \to \mathbb{R}$ and let x_0 be an accumulation point in E and $L \in \mathbb{R}$. We write $\lim_{x \to x_0} f(x) = L$ if

for every sequence $\{c_n\} \subseteq E$ with $c_n \neq x_0$. and $\lim_{n \to \infty} c_n = x_0$ then $\lim_{n \to \infty} (f(c_n)) = L$.

(δ-ε) and (sequential) definitions are equivalent.

Proof: Assume (δ-ε). Let $\{c_n\} \subseteq E$ with $c_n \neq x_0$ and $\lim_{n \to \infty} c_n = x_0$. By assumption, if $\varepsilon > 0$, $\exists \delta > 0$ s.t. $|f(x) - L| < \varepsilon$ when $|x - x_0| < \delta$. By convergence, there exists $N \in \mathbb{N}$ s.t. if $n \geq N$ then $|c_n - x_0| < \delta$. Thus $|f(c_n) - L| < \varepsilon$, and this is the sequential definition. ▨

Assume that $\lim_{x \to x_0} f(x) \neq L$. We must show that there exists a sequence $c_n \neq x_0$ with $\lim_{n \to \infty} c_n = x_0$ but $\lim_{n \to \infty} f(c_n) \neq L$. There must exist ε_0 s.t. $\forall \delta > 0$, there will be points satisfying $0 < |x - x_0| < \delta$ but $0 < |f(x) - L| \geq \varepsilon_0$. So, for $\delta_n = 1/n$ we can find c_n with $0 < |x_0 - c_n| < \delta_n$. and yet $|f(c_n) - L| \geq \varepsilon_0$. thus $\lim_{n \to \infty} f(c_n) \neq L$ as desired. ▨

Q.E.D.

Def'n: (open mapping). Let $f: E \to \mathbb{R}$ and let x_0 be an accumulation point of E and $L \in \mathbb{R}$. We write $\lim_{x \to x_0} f(x) = L$ if for every open set V with $L \in V$, there exists an open set U with $x_0 \in U$ s.t. if $x \in E \cap U \setminus \{x_0\}$ then $f(x) \in V$.

Proof: (δ-ε is equivalent to open mapping)

($\Rightarrow$) Assume (δ-ε).
Let V be an open set containing L, then L is an interior point of V. So, there exists $\varepsilon > 0$ s.t. $(L - \varepsilon, L + \varepsilon) \subseteq V$. Choose δ s.t. $|f(x) - L| < \varepsilon$ and $|x - x_0| < \delta$. Let $U = (x_0 - \delta, x_0 + \delta)$, so if $x \in U \cap E - \{x_0\}$, then $f(x) \in (L - \varepsilon, L + \varepsilon) \subseteq V$. ▨

($\Leftarrow$) Now assume the open mapping def'n. Let $\varepsilon > 0$ and set $V = (L - \varepsilon, L + \varepsilon)$. Then the open set U exists. Since $x_0 \in U$ and U is open, then x_0 is an interior point. So, $\exists \delta > 0$ s.t. $(x_0 - \delta, x_0 + \delta) \subseteq U$. This implies that $|f(x) - L| < \varepsilon$ if $|x - x_0| < \delta$. ▨

Q.E.D.

One-Sided Limits.

Let $f: E \to \mathbb{R}$ and x_0 is an accumulation point of $E \cap (x_0, \infty)$. We write $\lim_{x \to x_0^+} f(x) = L \in \mathbb{R}$ if

$\forall \varepsilon > 0, \exists \delta > 0$ s.t. $|f(x) - L| < \varepsilon$ when $|x - x_0| < \delta$.

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Section ...: 10/27

Def'n: Let $f: E \to \mathbb{R}$. We say that f is continuous on E if f is continuous for all $x \in E$.

Ex: Let $f:(0,1) \to \mathbb{R}$ where $f(x) = \begin{cases} 1/q, & \text{if } x = \frac{p}{q} \text{ (reduced)} \\ 0, & \text{if } x \in \mathbb{Q}. \end{cases}$

Let $c \in \mathbb{Q}$, so $c = p/q$, $p, q \in \mathbb{Z}^+$ So, $f(c) = \frac{1}{q}$.

$x_n = c + \frac{\pi}{n}$, $n \in \mathbb{N}$. Then each x_n is not in $\mathbb{Q}$, and thus $f(x_n) = 0$. Observe that $\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) \neq 1/q = f(c)$. Thus f is not continuous at c.

Now let $c \notin \mathbb{Q}$ and let $\varepsilon > 0$. By Archimedean Principle there exists $k \in \mathbb{N}$ with $1/k < \varepsilon$. Choose $\delta > 0$ s.t. if $p/q \in (c-\delta, c+\delta)$ then $\delta > k$. Let x satisfy $|x-c| < \delta$, so $x \in (c-\delta, c+\delta)$.

Case 1: Let $x \in \mathbb{Q}$, then p/q.
Then $|f(x) - f(c)| = |\frac{1}{q} - 0| = \frac{1}{q} < \frac{1}{k} < \varepsilon$.

Case 2: Let $x \notin \mathbb{Q}$, then $|f(x) - f(c)| = 0 < \varepsilon$.

In both cases, $|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon$.
$\therefore f$ is continuous on $\mathbb{R} \setminus \mathbb{Q}$ and discontinuous on $\mathbb{Q}$.

Section 5.5:

Theorem: Let $f, g: A \to \mathbb{R}$ and $c \in \mathbb{R}$. If f and g are continuous then cf, $f+g$, $f \cdot g$, f/g are all continuous. ($g \neq 0$).

Can be proven with $\lim$ laws.

Corollary: Every polynomial is continuous.

Proof: $f(x) = 1$ and $g(x) = x$ are continuous. ∎

Proposition: Let $f: A \to \mathbb{R}$ and $g: B \to \mathbb{R}$, where $f(A) \subseteq B$.

If f and g are continuous, then $g \circ f$ is continuous.

Proof: Let $\{x_n\} \subseteq A$ be a sequence with $\lim_{n\to\infty} x_n = x$, where $x \in A$.

$\therefore \lim_{n\to\infty} f(x_n) = f(x)$.

$\therefore \lim_{n\to\infty} g(f(x_n)) = g(f(x)) = g \circ f(x)$.

$\therefore g \circ f$ is continuous at x.

Q.E.D.

Section 5.6:

Def'n: Let $f: A \to \mathbb{R}$. We say f is <u>uniformly continuous</u> if $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $x, y \in A \wedge |x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$.

Ex: Let $f:(0,1) \to \mathbb{R}$ by $f: x \mapsto \frac{1}{x}$.

Proof: Set $\varepsilon = 1$ and let $\delta > 0$.
Let $x < \min\{\delta, 1\}$. Let $y = \frac{x}{2}$.

Then $|x - y| = |\frac{x}{2}| < \delta$.
However $|f(x) - f(y)| = |\frac{1}{x} - \frac{1}{y}|$

$= |\frac{1}{x} - \frac{2}{x}|$.

$= |\frac{1}{x}|$

$> 1 = \varepsilon$.

Thus f is not uniformly continuous.

Q.E.D.

Section 7.2: 11/03/25

Def'n: Let I be an interval with $f: I \to \mathbb{R}$, let $x_0 \in I$, the derivative of f at x_0 is:

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

If this lim exists, then f is differentiable at x_0.

Theorem: If f is differentiable then f is continuous (at x_0).

Proof: observe that $\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0)$ and $\lim_{x \to x_0} x - x_0 = 0$ both exist.

So, by the lim laws $\lim_{x \to x_0} f(x) - f(x_0)$

$$= \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \cdot x - x_0.$$

$$= \left(\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} \right) \cdot \lim_{x \to x_0} x - x_0$$

$$= f'(x_0) \cdot 0$$

$$= 0.$$

$\therefore \lim_{x \to x_0} f(x) = f(x_0)$. Thus f is continuous at x_0.

Q.E.D.

Section 7.3:

Power Rule: If $f(x) = x^n$, $n \in \mathbb{N}$, then $f'(x) = n x^{n-1}$.

Proof: HW (induction)

Lemma: If for every open interval I containing x_0 there exists $x \in I$, $x \neq x_0$, s.t. $f(x) = f(x_0)$ then either $f'(x_0)$ does not exist or else $f'(x_0) = 0$.

Proof: HW.

Theorem: Chain Rule. Let $f: A \to \mathbb{R}$ and $g: B \to \mathbb{R}$, $f(A) \subseteq B$. If f is differentiable at $c \in A$ and g is differentiable at $f(c)$ then $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$.

Proof: Let $\{x_n\}$ be a sequence with $x_n \neq x_0$ and $\lim_{n \to \infty} x_n = x_0$.

Let $S_n = \frac{g(f(x_n)) - g(f(x_0))}{x_n - x_0}$. We must prove that $\lim_{n \to \infty} S_n = g'(f(c)) \cdot f'(c)$

If $f(x_n) \neq f(x_0)$ then let $y_n = f(x_n)$ and $y_0 = f(x_0)$. Then:

$$S_n = \left(\frac{g(y_n) - g(y_0)}{y_n - y_0} \right) \cdot \left(\frac{f(x_n) - f(x_0)}{x_n - x_0} \right).$$

$$\Rightarrow \lim_{n \to \infty} S_n = \lim_{n \to \infty} \left(\frac{g(y_n) - g(y_0)}{y_n - y_0} \right) \cdot \lim_{n \to \infty} \left(\frac{f(x_n) - f(x_0)}{x_n - x_0} \right)$$

$$= g'(y_0) \cdot f'(x_0).$$

$$= g'(f(x_0)) \cdot f'(x_0). \quad \text{Hooray.}$$

Suppose $f(x_n) = f(x_0)$ $\forall n \geq N$.

Then by the lemma, $f'(x_0) = 0$. It follows that $\lim_{n \to \infty} S_n = 0 = g'(f(x_0)) \cdot f'(x_0)$.

Q.E.D.

Inverse function theorem: Suppose $f: A \to \mathbb{R}$ is differentiable on the interval I, $f'(x) \neq 0$. If f is bijective, then:

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad \forall x \in I.$$

Proof: Note that $f^{-1}(f(x)) = x$.

$$\Rightarrow (f^{-1})'(f(x)) \cdot f'(x) = 1$$

$$\Rightarrow (f^{-1})'(f(x)) = \frac{1}{f'(x)}.$$

Q.E.D.

Section 7.4:

Ex: Let $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0. \end{cases}$

Because $\sin\left(\frac{1}{x}\right) \leq 1$ it follows that $|f(x)| \leq x^2$.

Then $|f'(0)| = \lim_{x \to 0} \left| \frac{f(x)}{x} \right| \leq \lim_{x \to 0} x^2 = 0$.

$\Rightarrow |f'(0)| = f'(0) = 0$.

Let $x_n = 1/\pi_n$. Then $f'(x_n) = -\cos(\pi n) + \frac{2}{\pi n} \sin(n\pi)$.

$\therefore \lim_{n \to \infty} f'(x_n)$ DNE.

$f'(x)$ is defined for all x, but $f'(x)$ is not continuous.

Section 7.5: 11/5/25

Theorem: Let f be defined on the interval $I \subseteq \mathbb{R}$. If f has a local min or local max at $x_0 \in I$ and f is differentiable at x_0 then $f'(x_0) = 0$. x_0 is an interior point.

Proof: Suppose $f(x_0)$ is a local max WLOG. Then $\exists \delta > 0$ s.t.

$[x_0 - \delta, x_0 + \delta] \subseteq I$ and $\forall x \in [x_0 - \delta, x_0 + \delta]$, $f(x_0) \geq f(x)$.

- If $x \in (x_0, x_0 + \delta]$ then

$$\frac{f(x) - f(x_0)}{x - x_0} \leq 0.$$

- If $x \in [x_0 - \delta, x_0)$ then

$$\frac{f(x) - f(x_0)}{x - x_0} \geq 0.$$

$$f'(x_0) = \lim_{x \to x_0^+} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{x \to x_0^-} \frac{f(x) - f(x_0)}{x - x_0}.$$

Thus $0 \leq f'(x_0) \leq 0$.

So, $f'(x_0) = 0$.

Q.E.D.

Theorem: Intermediate Value Theorem.

Let f be continuous on $[a, b]$ and let $L \in \mathbb{R}$ be between $f(a)$ and $f(b)$. Then there exists $c \in [a, b]$ s.t. $f(c) = L$.

Proof: WLOG, assume that $f(a) \leq L \leq f(b)$.

Let $I_1 = [a, b]$. Given $I_n = [a_n, b_n]$. Let $m_n = \frac{a_n + b_n}{2}$

- If $f(m_n) > L$, let $I_{n+1} = [a_n, m_n]$
  else, let $I_{n+1} = [m_n, b_n]$.

Then $I_{n+1} \subseteq I_n$ and $b_n - a_n = \frac{b - a}{2^n} \to 0$.

Thus, by Cantor's intersection theorem, $\bigcap_{n=1}^{\infty} I_n = \{c\}$. with $\lim_{n \to \infty} a_n = c = \lim_{n \to \infty} b_n$.

But $f(a_n) \leq L \leq f(b_n)$

$$\Rightarrow \lim_{n \to \infty} f(a_n) \leq L \leq \lim_{n \to \infty} f(b_n)$$

$$\Rightarrow \lim_{n \to \infty} f(c) \leq L \leq \lim_{n \to \infty} f(c) \Rightarrow f(c) = L.$$

Q.E.D.

Section 7.6:

Rolle's Theorem: Let f be continuous on $[a, b]$ and differentiable on (a, b). If $f(a) = f(b)$ then there exists $c \in (a, b)$ s.t. $f'(c) = 0$.

Proof: If f is constant, pick any $c \in (a, b)$ and $f'(c) = 0$.

Assume f is not constant. Because $[a, b]$ is compact, $f([a, b])$ is compact. $f([a, b])$ is closed and bounded. Let $M = \max\{f[a, b]\}$, $m = \min\{f[a, b]\}$. Because f is not constant $m \neq M$. WLOG let $M > f(a)$. By the image there exists $c \in (a, b)$ with $f(c) = M$. Because M is a max and an interior point, $f'(c) = 0$.

Q.E.D.

Mean Value Theorem: Suppose f is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ s.t. $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Proof incoming.

...

Lipschitz Continuous on $[a,b] \iff f'$ is bounded.

Proof: Suppose lipschitz continuous on $[a,b]$.

$\therefore \forall x,y \in [a,b], \exists M$ s.t. $|f(x)-f(y)| \leq M|x-y|$.

Suppose $x=y$. $|f(x)-f(y)| = 0 \leq M|0| = 0$. ✓

Suppose $x < y$ without loss of generality.

We can form an interval

$[x,y]$

By MVT, $\exists c \in [x,y]$, s.t.

$\frac{|f(x)-f(y)|}{|x-y|} = f'(c) \leq M$.

We can now take $\lim_{x \to y} \frac{|f(x)-f(y)|}{|x-y|}$

$= \lim_{x \to y} f'(c) = f'(y) \leq M$.

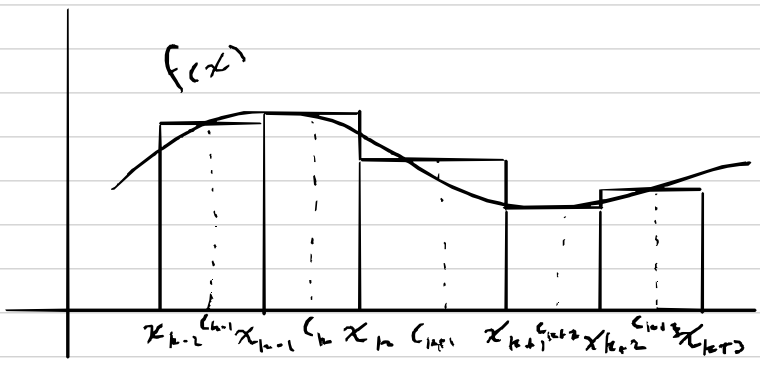
This works for $y \to x$ as well of course.

Thus, f' is bounded on (a,b).

($\Leftarrow$) Now, Suppose that f' is bounded on (a,b)

$\lim_{x \to z} \frac{|f(x)-f(z)|}{x-z} \leq M$.

Section 7.1: 11/17/25



Let $[a,b]$ be partitioned into subintervals $[x_{k-1}, x_k]$ with $c_k \in [x_{k-1}, x_k]$. Then

$\sum_{k=1}^{n} f(c_k)\cdot(x_k - x_{k-1})$ is the Riemann Sum.

Section 7.2:

Def'n: A partition P of $[a,b]$ is a finite list $\{x_0, x_1, \dots, x_n\}$ with $a = x_0 < x_1 < x_2 < \cdots < x_n = b$.

Given a subinterval $[x_{k-1}, x_k]$ let

- $m_k = \inf\{f(x) : x \in [x_{k-1}, x_k]\}$.
- $M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\}$.

- The lower sum with respect to P is

  $L(f, P) = \sum_{k=1}^{n} m_k (x_k - x_{k-1})$.

- The upper sum with respect to P is

  $U(f, P) = \sum_{k=1}^{n} M_k (x_k - x_{k-1})$.

Clearly $L(f,P) \le U(f,P)$.

Def'n: A partition Q is called a refinement of P if $P \subseteq Q$.

Lemma: $P \subseteq Q \Rightarrow L(f,P) \le L(f,Q) \wedge U(f,P) \ge U(f,Q)$.

Proof: Let $P = \{x_0, x_1, \dots, x_n\}$ and let Q be a refinement by just one point z. So, $Q = \{x_0, x_1, \dots, x_{k-1}, z, x_k, \dots, x_n\}$.

- $m_k' = \inf\{f(x) : x \in [z, x_k]\}$.

  $m_k'' = \inf\{f(x) : x \in [x_{k-1}, z]\}$.

Then $m_k(x_k - x_{k-1}) = m_k(x_k - z) + m_k(z - x_{k-1})$.

$\le m_k'(x_k - z) + m_k''(z - x_{k-1})$.

Thus $L(f,P) \le L(f,Q)$.

Inductive hypothesis.

Q.E.D.

Lemma: P_1, P_2 are partitions of $[a,b]$. Then

$L(f, P_1) \le U(f, P_2)$.

Proof: Let $Q = P_1 \cup P_2$. $P_1 \subseteq Q$ and $P_2 \subseteq Q$.

So, $L(f, P_1) \le L(f, Q) \le U(f, Q) \le U(f, P_2)$.

Q.E.D.

Def'n: Let $\mathcal{P}$ be the collection of all possible partitions.

- The upper integral of f is

  $U(f) = \inf\{U(f,P) : P \in \mathcal{P}\}$.

- The lower integral of f is

  $L(f) = \sup\{L(f,P) : \mathcal{P} \ni P\}$.

If f is bounded and $U(f) = L(f)$ then we say that f is Riemann integrable and

$$\int_a^b F(x)\,dx = U(f) = L(f).$$

Section 7: 11/19/25

Lemma: If f is bounded then $U(f) \geq L(f)$ is always true.

Proof: HW stuffs. Q.E.D?

- $P = \{x_0, x_1, \dots, x_n\}$ ← Partition.
- $m_k = \inf\{f(x) : x \in [x_{k-1}, x_k]\}$.
- $M_k = \sup\{f(x) : x \in [x_{k-1}, x_k]\}$.
- $L(f, P) = \sum_{k=1}^{n} m_k (x_k - x_{k-1})$.
- $U(f, P) = \sum_{k=1}^{n} M_k (x_k - x_{k-1})$.
- $L(f) = \sup\{L(f, P) : P \in \mathcal{P}\}$
- $U(f) = \inf\{U(f, P) : P \in \mathcal{P}\}$.

Theorem: (integrability criterion) A bounded function f is integrable on $[a, b]$ iff $\forall \varepsilon > 0$, $\exists P_\varepsilon \in \mathcal{P}([a,b])$ with $U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$.

Proof: Assume f is integrable, so $U(f) = L(f)$. Because $U(f)$ is a greatest lower bound there exists a partition P_1 with $U(f, P_1) < U(f) + \varepsilon/2$.

Similarly, $\exists P_2$ s.t. $L(f, P_2) > L(f) - \varepsilon/2$.

Let $P_\varepsilon = P_1 \cup P_2$.

$\Rightarrow U(f, P_1) + L(f) - \varepsilon/2 < U(f) + L(f, P_2) + \varepsilon/2$.
$\Rightarrow U(f, P_1) - L(f, P_2) < \varepsilon$.

Using P_ε, we have that

$U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$. ▨

($\Leftarrow$) If $\forall \varepsilon > 0$, $\exists P_\varepsilon$ s.t. $U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$. then $U(f) - L(f) \leq U(f, P_\varepsilon) - L(f, P_\varepsilon) < \varepsilon$.

Thus $U(f) - L(f) < \varepsilon$ $\forall \varepsilon > 0$.

So, $U(f) - L(f) = 0$. ▨

Q.E.D.

Ex: Let $f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$ on $[0,1]$.

↑ also denoted $1_{\mathbb{Q}}(x)$ or $\chi_{\mathbb{Q}}(x)$.

Let P be any partition of $[0,1]$. By density

- $m_k = 0$. So $L(f) = 0$.
- $M_k = 1$. So $U(f) = 1$.

So, $U(f) \neq L(f)$ and f is not integrable.

Theorem: If f is continuous on $[a,b]$ then f is integrable on $[a,b]$.

Proof: Because $[a,b]$ is compact, f is uniformly continuous on $[a,b]$.

So, if $\varepsilon > 0$, $\exists \delta > 0$ s.t. if

$|x - y| < \delta$ then $|f(x) - f(y)| < \frac{\varepsilon}{b-a}$.

Let $P \in \mathcal{P}$ where $x_k - x_{k-1} < \delta$.

Let $M_k = f(z_k)$ be the absolute max of f on $[x_{k-1}, x_k]$ and $m_k = f(y_k)$ be the absolute min on $[x_{k-1}, x_k]$.

$U(f, P) = \sum_{k=1}^{n} M_k (x_k - x_{k-1})$

$L(f, P) = \sum_{k=1}^{n} m_k (x_k - x_{k-1})$.

By uniform continuity,

$|z_k - y_k| < \delta \Rightarrow M_k - m_k < \frac{\varepsilon}{b-a}$.

Thus $U(f, P) - L(f, P) = \sum_{k=1}^{n} (M_k - m_k)(x_k - x_{k-1})$

$< \frac{\varepsilon}{b-a} \sum_{k=1}^{n} (x_k - x_{k-1})$

$= \varepsilon$.

Q.E.D.

Section 7.4: 12/03/25

Theorem: Assume $f:[a,b]\to\mathbb{R}$ is bounded and let $c\in(a,b)$. Then f is integrable on $[a,b]$ iff f is integrable on $[a,c]$ and $[c,b]$.

Then $\int_a^b f = \int_a^c f + \int_c^b f$.

Proof: If f is integrable on $[a,b]$ then $\exists P_\varepsilon \in \mathcal{P}$ s.t. $U(f,P_\varepsilon) - L(f,P_\varepsilon) < \varepsilon$. $(\varepsilon>0)$.

$P_1 = P_\varepsilon \cup \{c\} \cap [a,c]$.

$P_2 = P_\varepsilon \cup \{c\} \cap [c,b]$.

Then $U(f,P_1) - L(f,P_1) < \varepsilon$ and $U(f,P_2) - L(f,P_2) < \varepsilon$, so f is integrable on $[a,c]$ and $[c,b]$.

($\Leftarrow$) Let P_1 and P_2 be partitions of $[a,c]$ and $[c,b]$ s.t.

$U(f,P_1) - L(f,P_1) < \varepsilon/2$ and $U(f,P_2) - L(f,P_2) < \varepsilon/2$.

Then $P_1 \cup P_2$ satisfies.

$U(f,P) - L(f,P) < \varepsilon$.

Thus f is integrable with $P = P_1 \cup P_2$.

We know $\int_a^b f \le U(f,P) < \varepsilon + L(f,P)$.

$= L(f,P_1) + L(f,P_2) + \varepsilon$.

$\le \int_a^c f + \int_c^b f + \varepsilon$.

Thus $\int_a^c f + \int_c^b f + \varepsilon \ge \int_a^b f \quad \forall \varepsilon$.

$\Rightarrow \int_a^b f \le \int_a^c f + \int_c^b f$.

Now, we let $\int_a^c f + \int_c^b f \le U(f,P_1) + U(f,P_2)$.

$< L(f,P_1) + L(f,P_2) + \varepsilon$.

$= L(f,P) + \varepsilon$.

$\le \int_a^b f + \varepsilon$.

This holds $\forall \varepsilon$, and thus,

$\int_a^b f \ge \int_a^c f + \int_c^b f$ and $\int_a^b f \le \int_a^c f + \int_c^b f$.

It follows that:

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

Q.E.D.

Theorem: Let f and g be integrable on $[a,b]$. Then

i) $f+g$ is integrable and $\int_a^b f+g = \int_a^b f + \int_a^b g$.

ii) if $k\in\mathbb{R}$, then kf is integrable on $[a,b]$ and $\int kf = k\int f$.

iii) If $m \le f(x) \le M$ on $[a,b]$, then $m(b-a) \le \int_a^b f \le M(b-a)$.

iv) If $f(x) \le g(x)$, then $\int_a^b f \le \int_a^b g$.

v) If $|f|$ is integrable, then $\left|\int_a^b f\right| \le \int_a^b |f|$.

Proof: (iii) Let $P = \{a,b\}$.

$L(f,P) = \inf\{f(x) : x\in[a,b]\}\cdot(b-a)$

$\ge m(b-a)$.

$U(f,P) = \sup\{f(x) : x\in[a,b]\}\cdot(b-a)$

$\le M(b-a)$.

$L(f,P) \le \int_a^b f \le U(f,P)$.

$m(b-a) \le \int_a^b f \le M(b-a)$.

Q.E.D.

Proof: (iv) Let $h = g - f$, then $h \ge 0$ and

$\int_a^b h \ge 0$

$\Rightarrow \int_a^b g - f \ge 0 \Rightarrow \int_a^b g \ge \int_a^b f$.

Q.E.D.

Proof: (v) $-|f(x)| \le f(x) \le |f(x)|$.

$-\int_a^b |f(x)| \le \int_a^b f(x) \le \int_a^b |f(x)|$.

So, $\left|\int_a^b f\right| \le \int_a^b |f|$.

Q.E.D.

If $f_n \to f$, it is not always true that $\int f_n \to \int f$.

Let $f_n = \begin{cases} n, & 0 \le x \le 1/n \\ 0, & x = 0 \text{ or } x > 1/n. \end{cases}$

Observe that $\int_0 f_n = 1 \ \forall n$.

However, for $x\in[0,1]$ we have that $f_n(x) \to 0$ as $n\to\infty$.

Thus $\int \lim_{n\to\infty} f_n = 0 \neq 1$.

Theorem: Assume $f_n \to f$ uniformly and each f_n is integrable. Then

$$\lim_{n\to\infty} \int_a^b f_n = \int_a^b f.$$

Proof: $\left|\int_a^b f_n - \int_a^b f\right| = \left|\int_a^b f_n - f\right| \le \int_a^b |f_n - f|$ by iv.

Let $\varepsilon > 0$ because $f_n \to f$ uniformly converges there is $N\in\mathbb{N}$ s.t. if $n \ge N$, then

$|f_n(x) - f(x)| < \frac{\varepsilon}{b-a}$.

Thus:

$\int_a^b |f_n - f| \le \int_a^b \frac{\varepsilon}{b-a} = \varepsilon$.

Q.E.D.

Section 7.5: 12/8/25

Fundamental Theorem of Calculus:

i) If $f:[a,b]\to\mathbb{R}$ is integrable and $F:[a,b]\to\mathbb{R}$ satisfies $F'(x)=f(x)$ for $x\in[a,b]$, then:

$$\int_a^b f = F(b)-F(a).$$

ii) Let $g:[a,b]\to\mathbb{R}$ be integrable and for $x\in[a,b]$, define:

$$G(x)=\int_a^x g$$

Then G is continuous and if g is continuous then G is differentiable and $G'(x)=g(x)\ \forall x\in[a,b]$.

Proof: (i) Let $P=\{x_0=a, x_1, \dots, x_n=b\}$ be a partition of $[a,b]$.

Apply the Mean-Value-Theorem to F on $[x_{k-1},x_k]$ of P. So, $\exists t_k\in[x_{k-1},x_k]$ s.t.

$$\begin{aligned}F(x_k)-F(x_{k-1}) &= F'(t_k)(x_k-x_{k-1})\\ &= f(t_k)(x_k-x_{k-1}).\end{aligned}$$

Now $m_k\le f(t_k)\le M_k$. Thus:

$$L(f,P)\le\sum_{k=1}^{n}[F(x_k)-F(x_{k-1})]\le U(f,P).$$

$$\Rightarrow L(f,P)\le F(b)-F(a)\le U(f,P).$$

Thus $F(b)-F(a)$ is a lower bound for:

$$\{U(f,P): P\in\mathcal{P}\}.$$

Thus, $F(b)-F(a)\le U(f)$.

Similarly, $F(b)-F(a)$ is an upper bound for:

$$\{L(f,P): P\in\mathcal{P}\}.$$

Thus, $F(b)-F(a)\ge L(f)$. Because f is integrable, $U(f)=L(f)$ and thus it follows that

$$F(b)-F(a)=\int_a^b f. \quad \blacksquare$$

ii) Let $x,y\in[a,b]$. Assume WLOG $x>y$. Let M be a bound for g.

$$|G(x)-G(y)|=\left|\int_a^x g-\int_a^y g\right|=\left|\int_y^x g\right|$$

$$\le\int_y^x |g|\le M|x-y|.$$

Thus g is lipschitz continuous, and thus g is uniformly continuous.

Now assume g is continuous at $c\in[a,b]$. Then:

$$\lim_{x\to c}\frac{G(x)-G(c)}{x-c}$$

$$=\lim_{x\to c}\frac{1}{x-c}\int_c^x g(t)\,dt.$$

Let $\varepsilon>0$. We want $\delta>0$ s.t. if $|x-c|<\delta$ then

$$\left|\frac{1}{x-c}\int_c^x g(t)\,dt-g(c)\right|<\varepsilon.$$

By continuity of g there exists $\delta>0$ s.t. if

$$|t-c|<\delta \text{ then } |g(t)-g(c)|<\varepsilon,$$

where $|t-c|\le|x-c|$. So,

$$\left|\frac{1}{x-c}\int_c^x g(t)\,dt-g(c)\right|.$$

$$\Rightarrow\left|\frac{1}{x-c}\int_c^x [g(t)-g(c)]\,dt\right|.$$

$$\le\frac{1}{x-c}\int_c^x |g(t)-g(c)|\,dt.$$

$$=\frac{1}{x-c}\int_c^x \varepsilon\,dt=\varepsilon.$$ As we assumed that

$|t-c|<\delta\Rightarrow|g(t)-g(c)|<\varepsilon$ by continuity.

Thus, $G'(c)=\lim_{x\to c}\frac{G(x)-G(c)}{x-c}=g(c)$. $\blacksquare$

Q.E.D.